

Q1) MCQ:

1- A data structure that maps terms back to the parts of a document in which they occur is called an (select the best answer):

- a) Postings list
- b) Incidence Matrix
- c) Dictionary
- d) Inverted Index

2- In information retrieval, the number of times that a word or term occurs in a document is called the:

- a) Proximity Operator
- b) Vocabulary Lexicon
- c) Term Frequency
- d) Indexing Granularity

3- The common words which would appear to be from the index vocabulary are called:

- a) Stop Words
- b) Lemmatized Words
- c) Tokens
- d) Stemmed Terms

4- In IR, a metric that equal the log of N divided by the document frequency where N is the total number of documents in a collection is

- a) document frequency
- b) collection frequency
- c) tf-idf weight
- d) inverse document frequency

5) In information retrieval, A measure of similarity between two vectors which is determined by measuring the angle between them is called:

- a) cosine similarity
- b) vector scoring
- c) sin similarity
- d) vector similarity

6) The web crawler starts with a set of as initial start.

- a) Web pages
- b) Seeds
- c) Topics
- d) Queries

7) From query refinement techniques

- a) query expansion
- b) relevance feedback
- c) query suggestion
- d) All of the mentioned

8) During process, documents are retrieved in sorted order according to a score computing using the document representation.

- a) Ranking
- b) Pre-processing
- c) Text transforming
- d) Text acquisition

9) Web site administrators can control web crawlers from access their webpages through

- a) Sitemap
- b) Home page
- c) Robots file
- d) Restrictions rules

10) The task of assignment of numerical values to terms that represent their importance in a document in order is called

- a) Term selection
- b) Document representation
- c) Term weighting
- d) Term indexing

رقم 5 في امتحان 2021 هتبقى all of the mentioned

Q2) True or False:

1) Recall is the number of correct results divided by the number of all returned results

- a) True **b) False**

2) System A achieves a MAP of 0.33 and System B achieves a MAP of 0.79, this means that System A is significantly better than System B.

- a) True **b) False**

3) For vector space model, only cosine similarity measure could be used.

- a) True **b) False**

4) When applying the Stemming preprocessing step, we increase the size of vocabulary.

- a) True **b) False**

5) To get perfect recall, the system should steal documents.

- a) True **b) False**

6) Search engine design should balance between two factors, performance and effectiveness.

- a) True** b) False

7) The similarity function should be larger for documents that better satisfy a query's information need.

- a) True** b) False

8) Term weighting is a procedure that takes place during the text indexing process in order to assess the value of each term to the document.

a) True b) False

9) In IR, a process to efficiently intersect posting lists to be able to quickly find documents that contain both terms is referred to as merging postings lists.

a) True b) False

10) During index creation, each document in the search collection is given a unique number.

a) True b) False

Q3) Give short answers for the following questions.

1) In inverted index compression, what part of an inverted index is usually compressed?

- The posting lists within the inverted index.

2) List three models of retrieval models?

1- Simple models: – Boolean retrieval – Vector Space model

2- Probabilistic Models: – Language models – BM25 .

3- Combining evidence: – Inference networks – Learning to Rank.

3) What is the difference between the generic crawlers and focused crawlers?

Generic crawlers: Grabs all documents & links related to a topic, retrieve a large number of pages from different fields.

Focused crawlers: download pages about a specific topic, Used by vertical search applications.

4) Mention at least one of the Open source search engines that are important for research and development.

OpenAIRE : is a global open access repository that indexes scientific research publications.

- is: Freely Available & Comprehensive

Lucene: Is a high-performance, full-featured text search engine library written in Java.

- is: Scalable & Extensible

5) What is the Deep web

Deep web: parts of the world wide web whose contents aren't indexed by standard web search engines.

- Often hidden behind login forms, **include**: Web mail & Online banking.